MPA-RES 20/27Z
Homework 2
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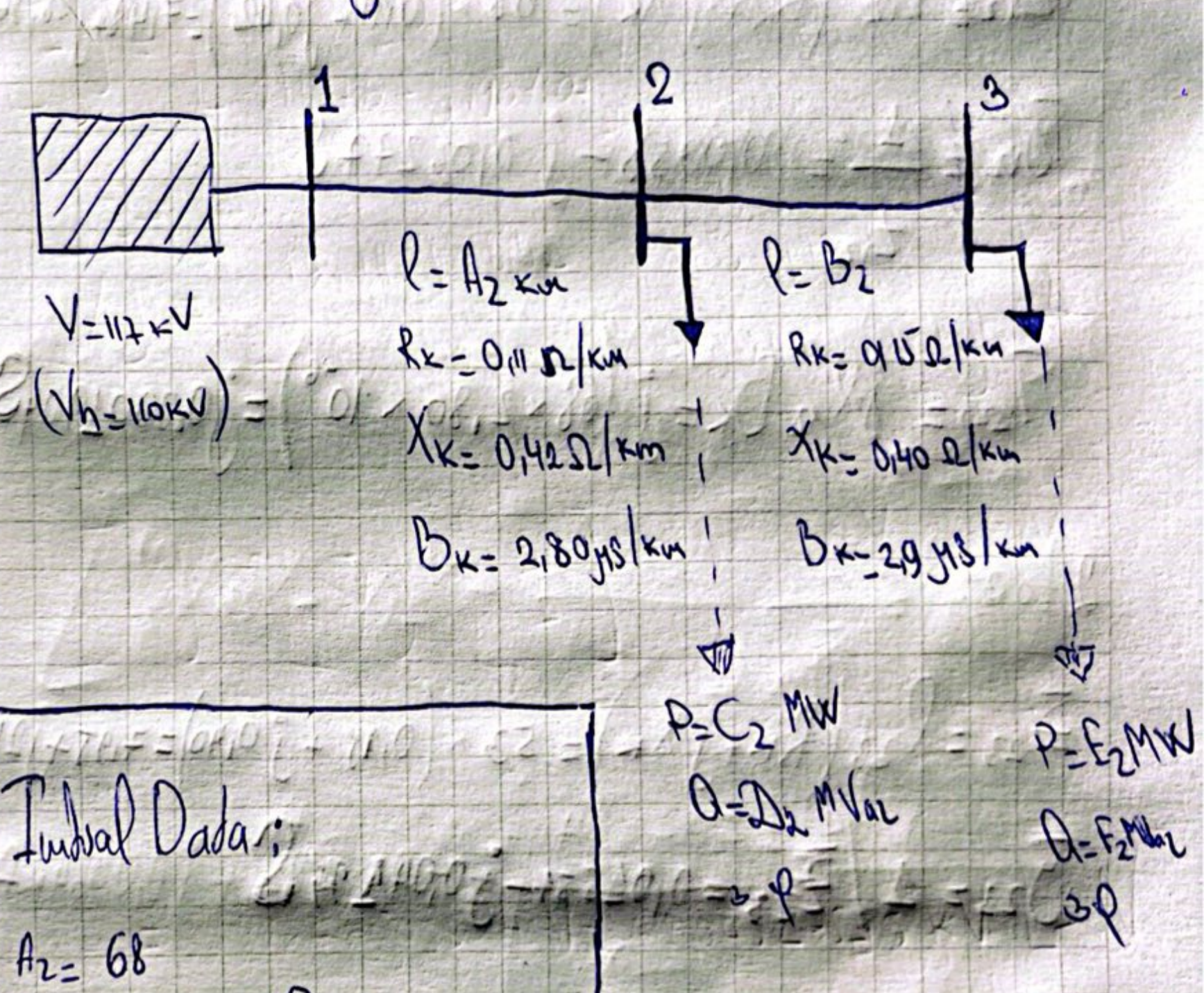

Initial Data:

$A_2=68$
$D_2=4$
$B_2=53$
$E_2=34$
$C_2=57$
$F_2=3$

Line Parameters:

1 to 2:

$\bar{Z}_{12}=l_{12}(R_K+jX_K)=68\times(0,11+j0,42)=7,48+j28,56$

$\bar{y}_{12}=\frac{1}{\bar{Z}_{12}}=0,0058-j\,0,03277\,S$

$\bar{Y}_{120}=j l_{12}B_K=j(68\times2,80\times10^{-6})=j0,0001904\,S$

2 to 3:

$\bar{Z}_{23}=l_{23}(R_K+jX_K)=53\times(0,15+j0,40)=7,95+j21,2\,\Omega$

$\bar{y}_{23}=1/\bar{Z}_{23}=0,01551-j0,04135\,S$

$\bar{Y}_{230}=j\cdot l_{23}\cdot B_K=j(53\times2,9\times10^{-6})=j0,0001537\,S$

Admittance Matrix:

$\bar{Y}_{22}=\bar{y}_{12}+\bar{y}_{23}+\frac{\bar{Y}_{120}}{2}+\frac{\bar{Y}_{230}}{2}=0,02409-j0,07395\,S$

$\bar{Y}_{33}=$ [redacted] $\bar{y}_{23}+\frac{\bar{Y}_{230}}{2}=0,01551-j0,04128\,S$

$\bar{Y}_{21}=-\bar{y}_{12}=-0,00858+j0,03277\,S$

$\bar{Y}_{23}=-\bar{y}_{23}=-0,01551+j0,04135\,S$

$\bar{Y}_{31}=\bar{Y}_{13}=0\,S$

For efficient solving we first calculate inverse diagonals:

$\frac{1}{\bar{Y}_{22}}=3,9826+j12,2255\,\Omega$

$\frac{1}{\bar{Y}_{33}}=7,9760+j21,2297\,\Omega$

Gauss-Seidel Iterations:

$$\begin{bmatrix} V_1=117+j0\,kV \\ V_2^{(0)}=110+j0\,kV \\ V_3^{(0)}=110+j0\,kV \end{bmatrix}$$

First Iteration (k=1)

Bus 2 →

$V_2^{(1)}=\frac{1}{\bar{Y}_{22}}\left[\left(\frac{P_2-jQ_2}{V_2^{(0)*}}\right)-Y_{21}V_1-Y_{23}V_3^{(0)}\right]$

$\frac{P_2-jQ_2}{V_2^{(0)*}}=\left(\frac{-57-j4}{110}\right)^*=-0,5182+j0,0364\,kA$

$\bar{Y}_{21}V_1=(-0,00858+j0,03277)\times117=-1,0041+j3,8337\,kA$

$\bar{Y}_{23}V_3^{(0)}=(-0,0151+j0,04135)\times110=-1,7059+j4,5490\,kA$

$I_2^{(1)}=(-0,5182+j0,0364)-(-1,0041+j3,8337)-(-1,7059+j4,5490)=2,1917-j8,3463\,kA$

$V_2^{(1)}=\frac{1}{\bar{Y}_{22}}\cdot I_2^{(1)}=(3,9826+j12,2255)\times(2,1917-j8,3463)=$

$=110,7664-j6,4446\,kV$

$$\boxed{V_2^{(1)}=110,7664-j6,4446\,kV}$$

Bus 3 Voltage:

$V_3^{(1)}=\frac{1}{\bar{Y}_{33}}\left[\left(\frac{P_3-jQ_3}{(V_3^{(0)})^*}\right)-Y_{32}V_2^{(1)}\right]$

$\frac{P_3-jQ_3}{(V_3^{(0)})^*}=\left(\frac{-34-j3}{110}\right)^*=-0,3091+j0,0273\,kA$

[redacted] $\bar{Y}_{32}V_2^{(1)}=(-0,0151+j0,4135)\times(110,7664-j6,4446)=-1,412+j4,6806\,kA$

$V_3^{(1)}=(7,9760+j21,2297)\times(1,1421-j4,6533)=$

$=107,8989-j12,8674\,kV$

$$\boxed{V_3^{(1)}=107-j12,8674\,kV}$$

Second Iteration (k=2)

Bus Voltage 2 →

$V_2^{(2)}=\frac{1}{\bar{Y}_{22}}\left[\left(\frac{P_2-jQ_2}{V_2^{(1)*}}\right)-\bar{Y}_{21}V_1-\bar{Y}_{23}V_3^{(1)}\right]$

$\frac{P_2-jQ_2}{V_2^{(1)*}}=\left(\frac{-57-j4}{110,7664-j6,4446}\right)^*=-0,5108+j0,0658\,kA$

$\bar{Y}_{21}V_1$ → remains same as before calculations

$\bar{Y}_{23}V_3^{(1)}=(-0,0151+j0,04135)\times(107,8989-j12,8674)=$

$=-1,1412+j4,6617\,kA$

$I_2^{(2)}=(-0,5108+j0,0658)-(-1,0041+j3,8337)-(-1,1412+j4,6617)=1,6345-j8,4295\,kA$

$V_2^{(2)}=(3,9826+j12,2255)\times(1,6345-j8,4295)=$

$109,5641-j13,5892\,kV$

Bus 3 Voltage →

$V_3^{(2)}=\frac{1}{\bar{Y}_{33}}\left[\frac{P_3-jQ_3}{V_3^{(1)*}}-\bar{Y}_{32}V_2^{(2)}\right]$

$\frac{P_3-jQ_3}{V_3^{(1)*}}=\left(\frac{-34-j3}{107,8989-j12,8674}\right)^*=$

$=-0,3074+j0,0645\,kA$

$Y_{32}V_2^{(2)}=(-0,0151+j0,04135)\times(109,5641-j13,5892)$

$=-1,1371+j4,7417\,kA$

$I_3^{(2)}=(-0,3074+j0,0645)-(-1,1371+j4,7417)=$

$=0,8297-j4,6772\,kA$

$V_3^{(2)}=(7,9760+j21,2297)\times(0,8297-j4,6772)=$

$=105,9141-j19,6908\,kV$

Accuracy:

$\Delta V_3=V_3^{(2)}-V_3^{(1)}=(105,9141-j19,6908)-(107,8989-j12,8674)$

$\Delta V_3=-1,9848-j6,8234\,kV$

Absolute accuracy:

$|\Delta V_3|=\sqrt{(-1,9848)^2+(-6,8234)^2}=7,106\,kV$

Relative accuracy:

$\frac{|\Delta V_3|}{V_n}\times100\%=\frac{7,106}{110}=6,46\,\%$

It is really high for steady state load flow.

CamScanner